NYU

From 2D Pose Understanding to 3D Human Recovery
OpenPose, HRNet, and HMR

Yiqiao Liu

Taijia Liang

Intro to CV

CSCI-GA.3033-076

How has human pose estimation progressed from 2D pose estimation to 3D human recovery, and what roles have representation design, network architecture, and supervision strategy played in this transition?

Research Question

Relationship

Single 2D Posture Estimation

- Higher Precision — High-Resolution Network (HRNet)
- Multi-people target — OpenPose
- 3D Recovery — Human Mesh Recovery (HMR)

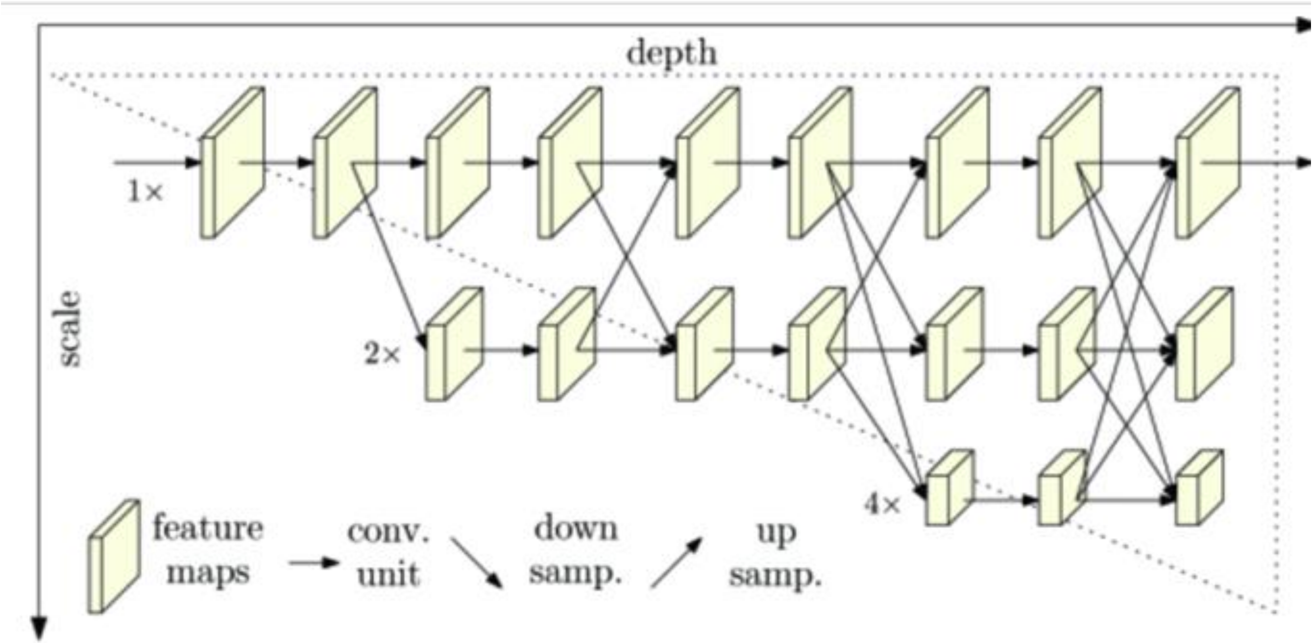
Sun et al. · CVPR 2019

HRNet

	Traditional Method	HRNet
01	perform downsampling first and then upsampling	keeps a high-resolution branch throughout the whole network
02	serial architectures, where features are progressively compressed from high resolution to low resolution	parallel multi-resolution branches to process different scales simultaneously.
03	recover resolution only in later decoding stages	repeatedly exchanges information across branches through repeated multi-scale fusion
	lose fine-grained spatial details during downsampling	

Sun et al. · CVPR 2019

HRNet-Methodology



Cao et al. · TPAMI 2019

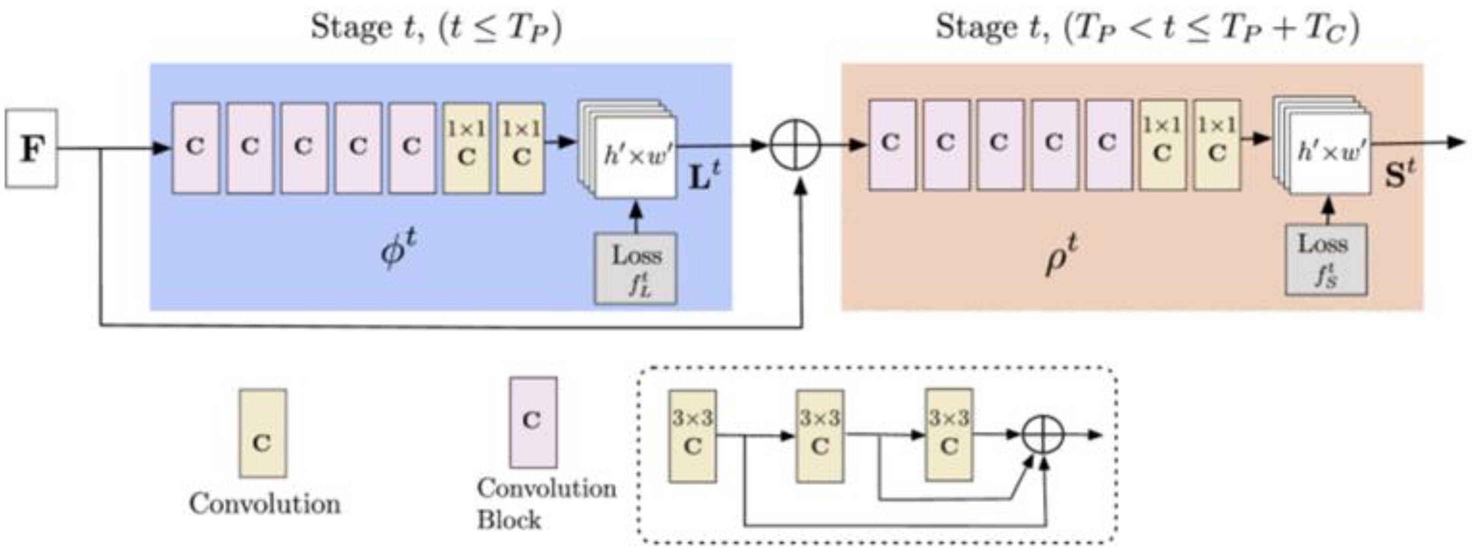
OpenPose

	Top-Down	Bottom-Up
01	Repeat single-person pose estimation	real-time multi-person 2D pose estimation
02	rely on keypoint heatmaps alone	confidence maps and Part Affinity Fields (PAFs)
03	Fully Connected Graph OutPut	tree-structured human skeleton
	High Cost + Low accuracy on complex spatial interaction	

NYU

Cao et al. · TPAMI 2019

OpenPose-Methodology



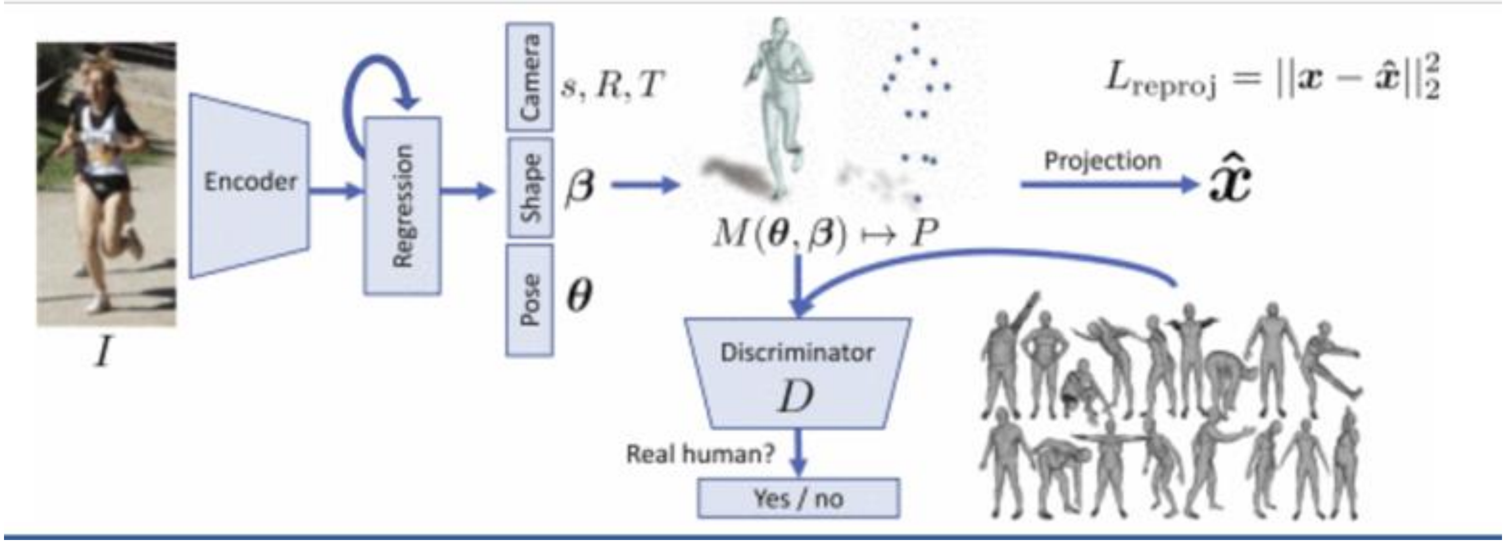
Kanazawa et al. · CVPR 2018

HMR

	Traditional	HMR
01	Just 3D joints	outputs a full 3D human mesh through the SMPL body model.
02	follow a two-stage process, first estimating 2D keypoints and then lifting them to 3D	end-to-end framework that directly regresses 3D model parameters from image pixels.
03	depend more heavily on explicit 2D-to-3D mapping or paired 3D annotations	combines 2D reprojection loss with an adversarial prior to reduce reliance on dense paired 3D supervision.
	discard contour, orientation, and shape cues + more vulnerable to depth ambiguity	

Kanazawa et al. · CVPR 2018

HMR-Methodology



Demo Workflow

One detector feeds 2D diagnostics, HMR2 mesh recovery, and PHALP video tracking.

IMAGE

RGB photos
4 test images

→

ViTDet-H
person boxes

→

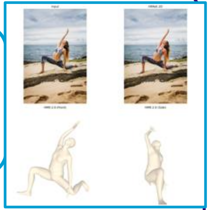
HRNet-W48
2D overlay

→

HMR 2.0
SMPL mesh

→

Showcase
quadplot + GIF



VIDEO

Video clip
8.6 s video

→

PHALP
detect + track

→

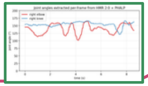
HMR / frame
per-frame SMPL

→

Track tensor
T×45×3 joints

→

Analytics
trajectory + angles



2D Keypoints

Stage 1 turns RGB images into reliable 2D body evidence.

01 **RGB image** — *input photo*

→

02 **ViTDet-H** — *person boxes score > 0.5*

→

03 **HRNet-W48** — *top-down 2D keypoints*

→

04 **COCO-17** — *17 joints + confidence*

→

05 **Artifacts** — *overlay JPG NPZ cache*

2D Showcase

HRNet overlays expose both easy and failure-prone pose evidence.

Standing

Complex pose

Occluded

Multi-person

NYU

3D Mesh

Stage 2 reuses detector boxes, then regresses SMPL in one forward pass.

01 **ViTDet-H boxes** → *shared detector output*

02 **HMR2 crop** → *person crop 224×224*

03 **ViT backbone** → *single forward pass*

04 **(θ, β, cam)** → *72 + 10 + 3 parameters*

05 **SMPL mesh** *6890 vertices front/side/GIF*

Important correction

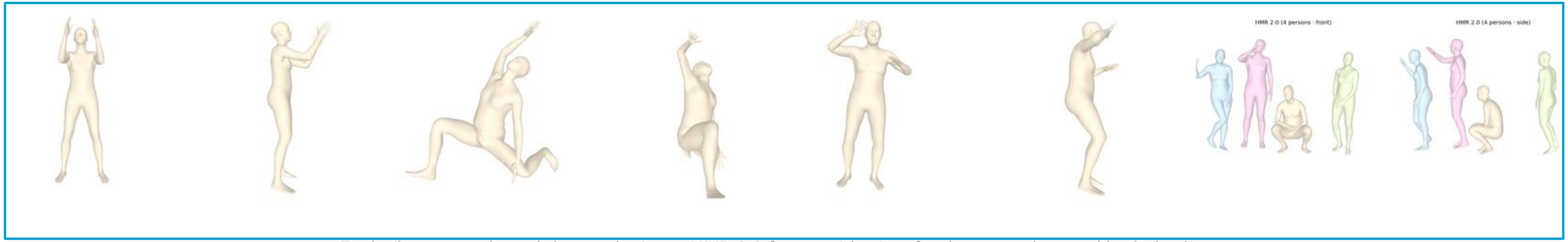
HRNet keypoints are visual evidence and validation. HMR2 does not consume the HRNet overlay in this demo.

Saved outputs

demo/results/hmr2_meshes.npz → demo/showcase/quadplot_*.png + rotation_*.gif

Mesh Showcase

Mesh views make pose visible outside the original camera.



Each tile uses real quadplot mesh views: HMR 2.0 front + side view for the same detected body/bodies.

4D Tracking

Stage 3 adds identity over time: per-frame HMR becomes a 4D track.

01 Video frames
tennis.mp4
8.6 s clip

→

02 PHALP loop
detect + crop
per frame

→

03 HMR 2.0
SMPL + joints
per detection

→

04 Track matching
pose + appearance
+ location

→

05 4D output
per-track
T×45×3 joints

Presentation phrasing

The valuable output is not just an overlay video; it is a structured time-series of SMPL pose and 3D joints.

Command path

python -m phalp.track video.source=demo/test_videos/tennis.mp4 → demo/results/tennis_phalp

Use this slide to explain what changes from single images to video: identity consistency plus time-indexed joints.

4D Showcase

4D Showcase

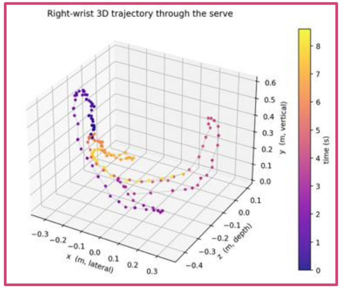
PHALP preserves identity and exports motion measurements.

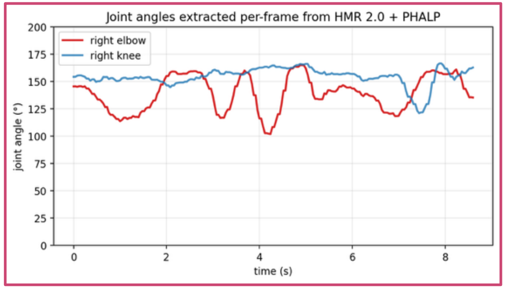
t = 0.5s

t = 2.0s

t = 4.0s

t = 6.0s





Showcase assets: PHALP overlay frames, right-wrist 3D trajectory, and elbow/knee angles from the per-track joint tensor.

Conclusion

What the demo proves, and where it still breaks.

01 TAKE-AWAYS

- 2D is the supervision and diagnostic layer for 3D.
- HMR2 makes single-image mesh recovery fast enough for a live demo.
- Adding time turns meshes into reusable motion data.

02 LIMITATIONS

- Per-frame video results still jitter and can drift in global scale.
- Occlusion remains the hardest image case.
- SMPL captures body pose/shape but not clothing, hair, or soft tissue.

03 NEXT STEPS

- Temporal smoothers such as VIBE/SLAHMR can reduce jitter.
- Human foundation models can improve robustness across domains.
- Neural or Gaussian avatars can represent surfaces beyond SMPL.

Thank you.